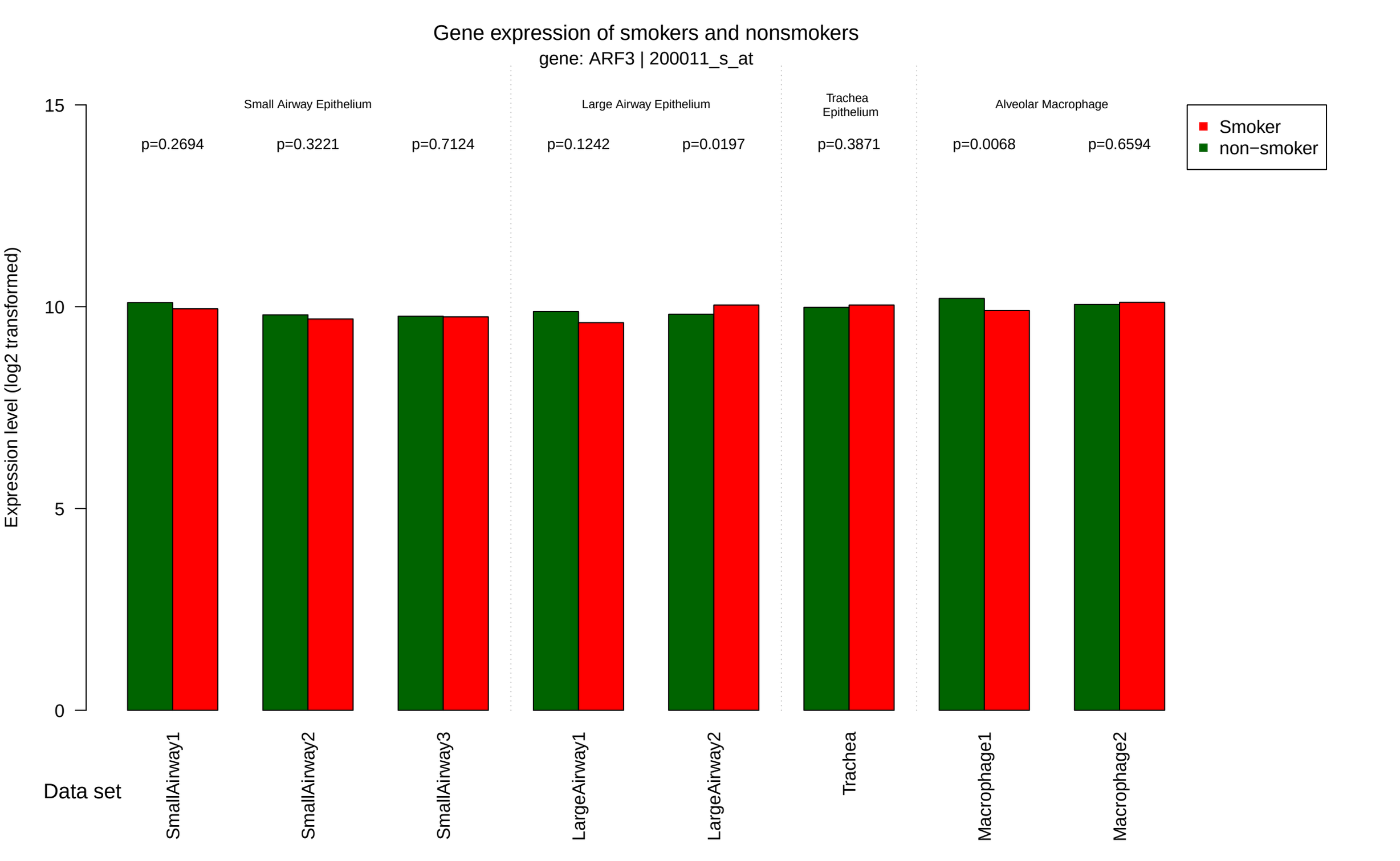
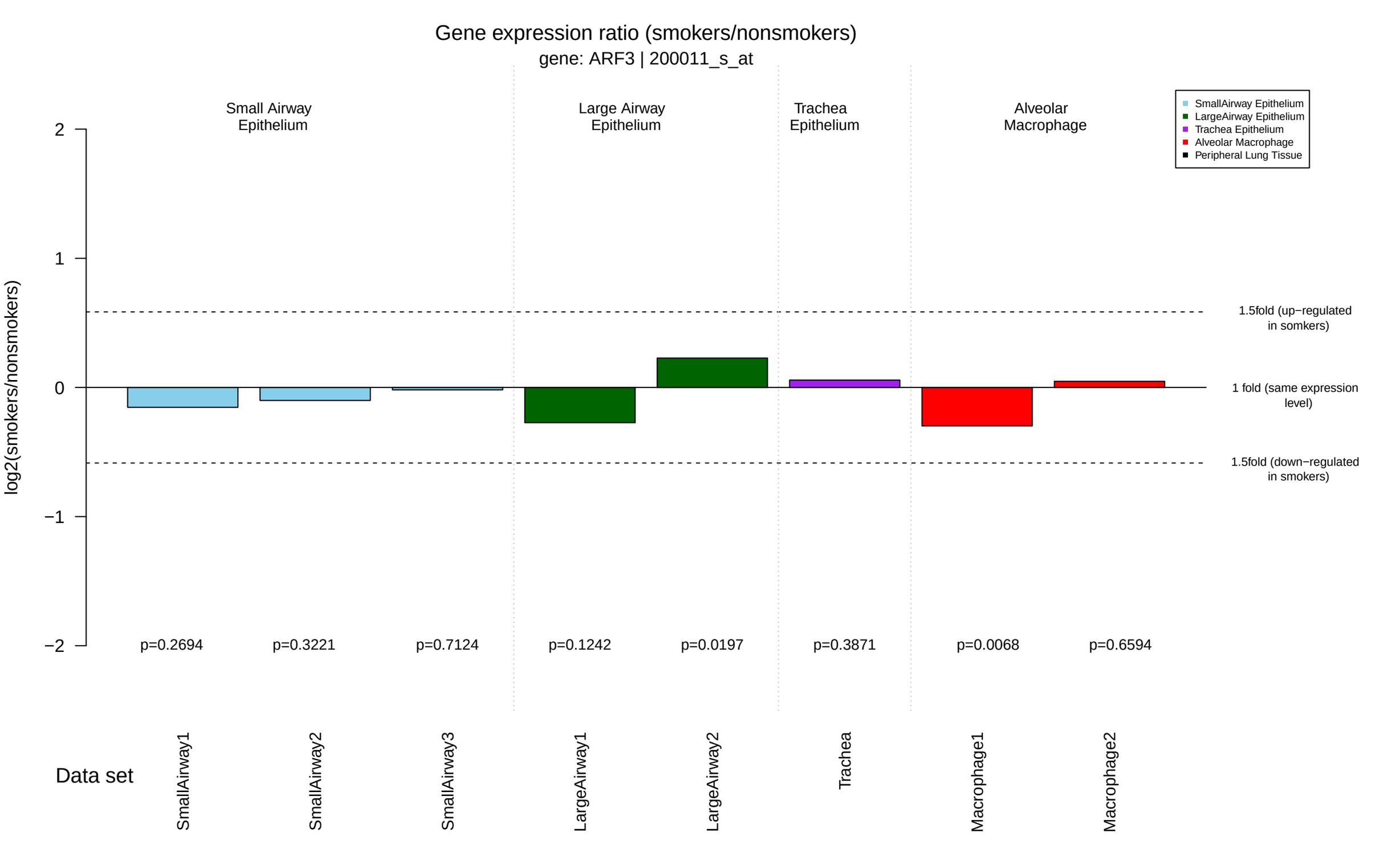
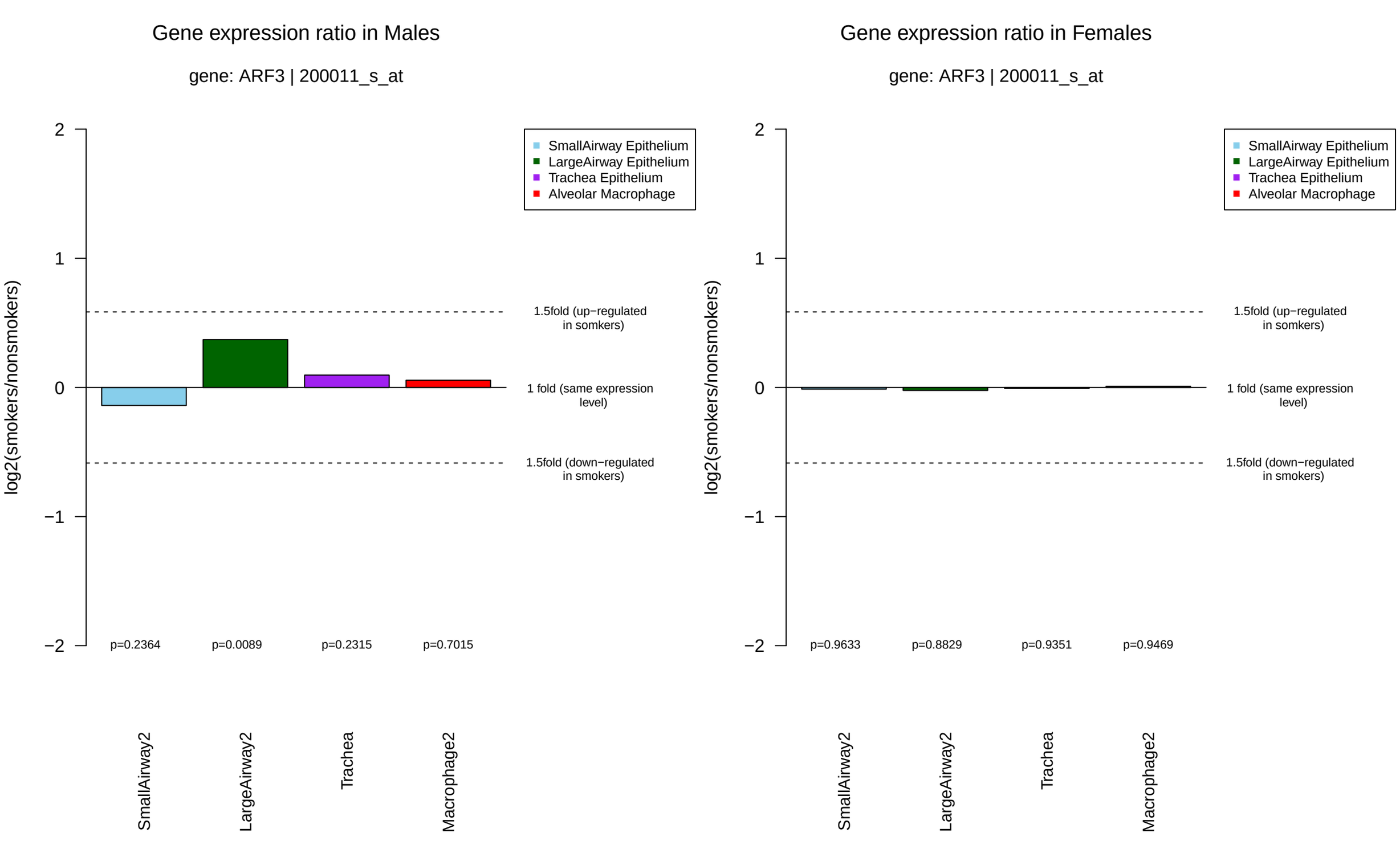
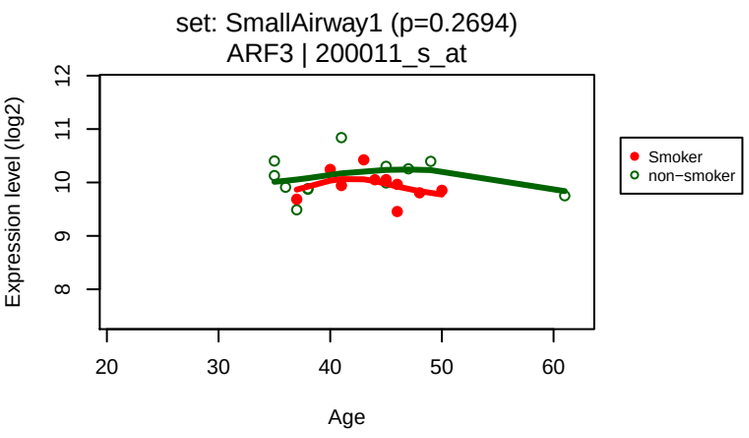
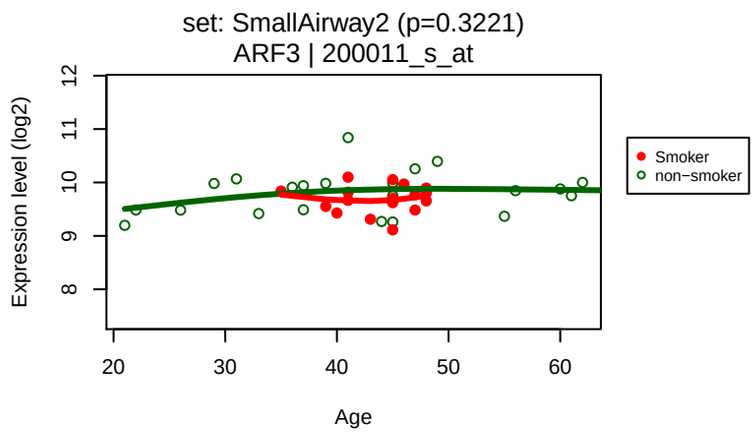
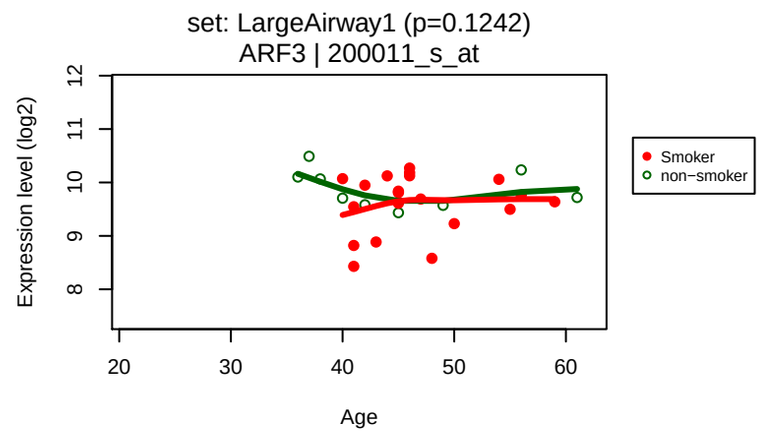

Gene expression of smokers and nonsmokers
gene: ARF3 | 200011_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2694
p=0.3221
p=0.7124
p=0.1242
p=0.0197
p=0.3871
p=0.0068
p=0.6594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

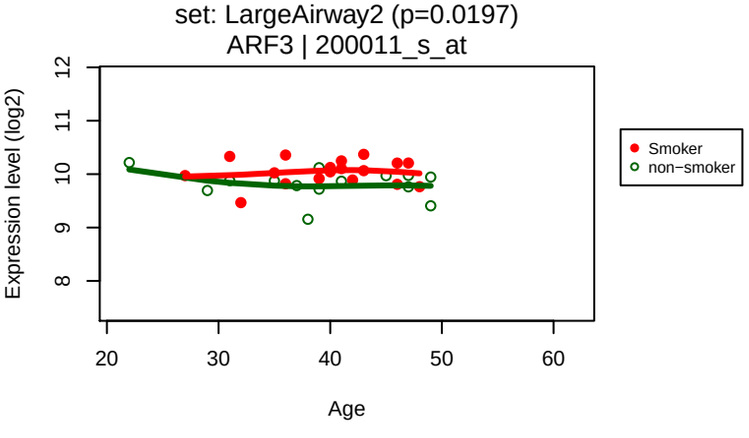

Gene expression ratio (smokers/nonsmokers)
gene: ARF3 | 200011_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2694
p=0.3221
p=0.7124
p=0.1242
p=0.0197
p=0.3871
p=0.0068
p=0.6594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

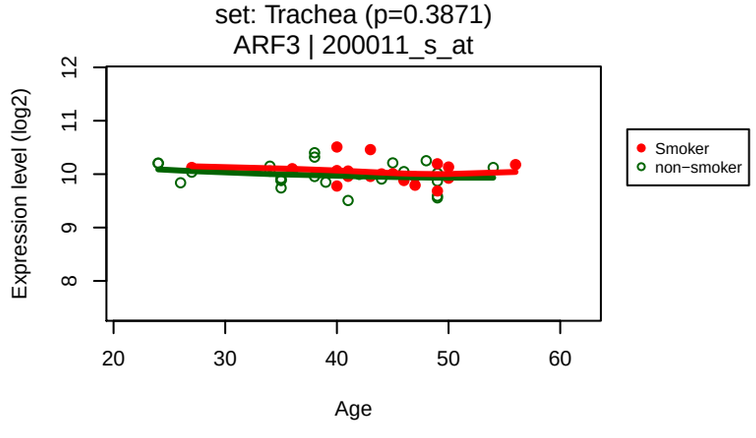

Gene expression ratio in Males
gene: ARF3 | 200011_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2364
p=0.0089
p=0.2315
p=0.7015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARF3 | 200011_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9633
p=0.8829
p=0.9351
p=0.9469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

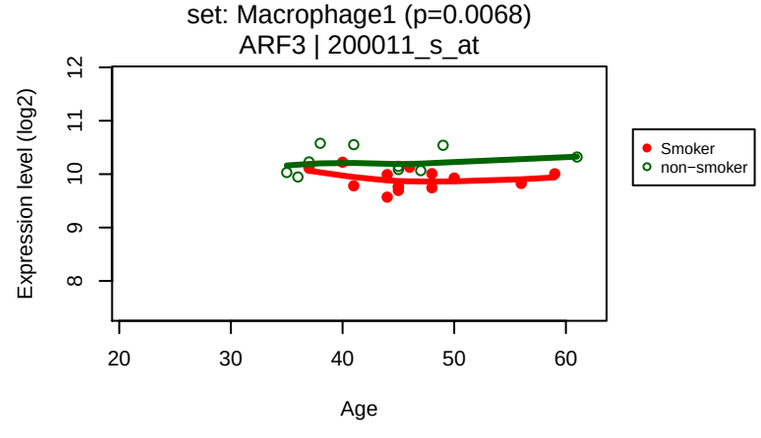

set: SmallAirway1 (p=0.2694)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker

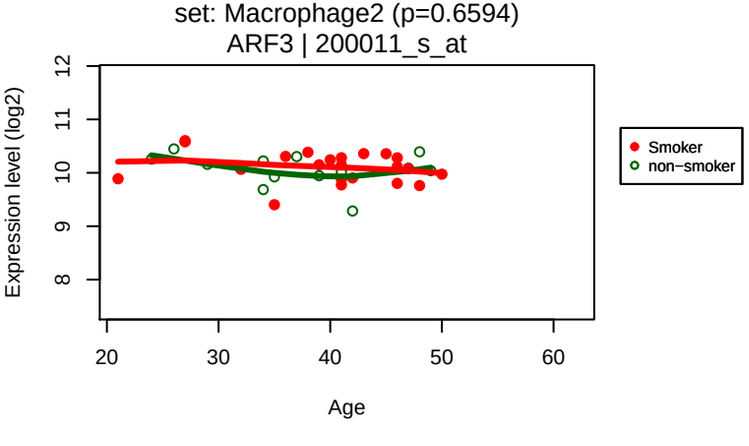

set: SmallAirway2 (p=0.3221)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker

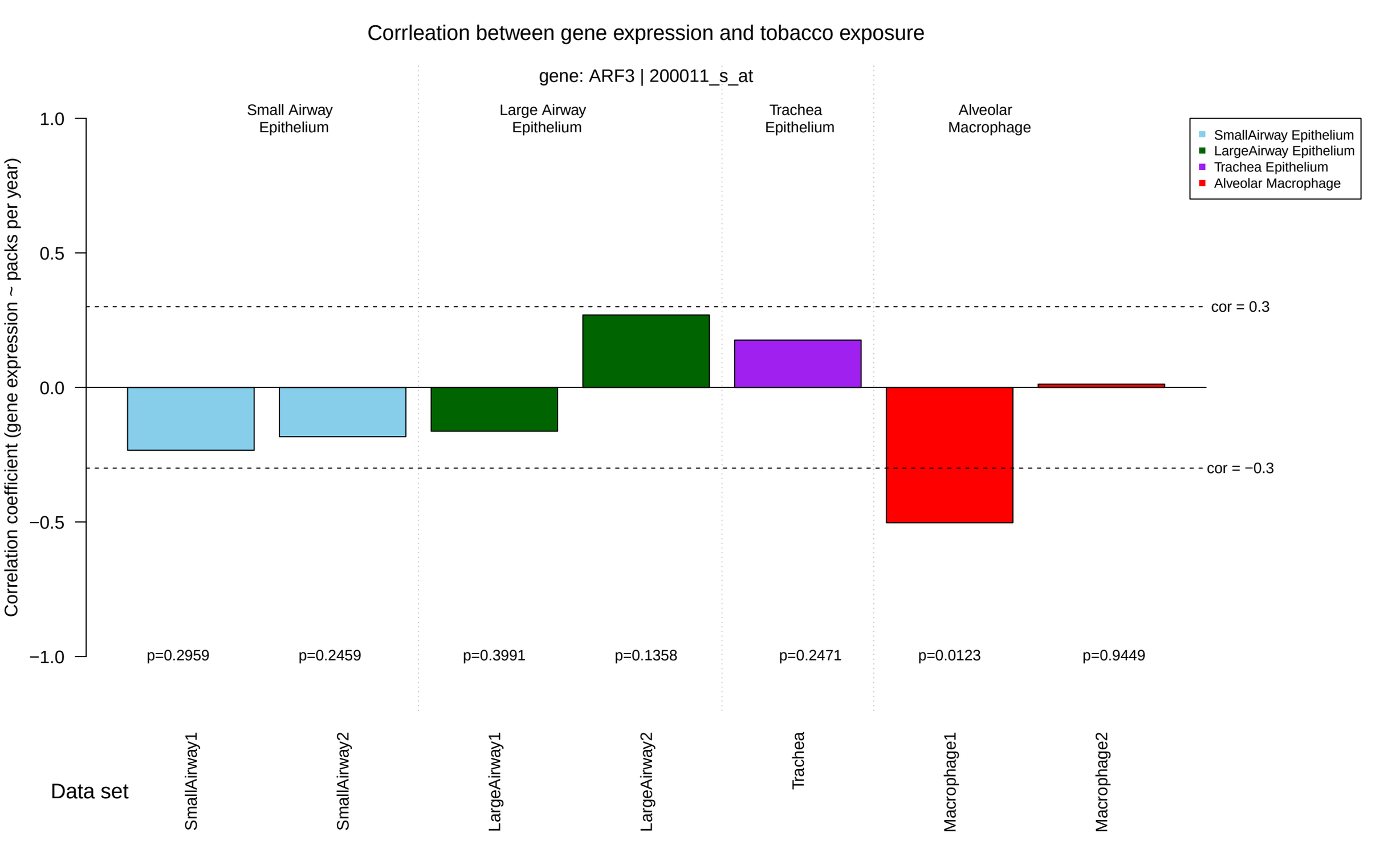

set: LargeAirway1 (p=0.1242)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0197)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3871)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0068)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6594)
ARF3 | 200011_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARF3 | 200011_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2959
p=0.2459
p=0.3991
p=0.1358
p=0.2471
p=0.0123
p=0.9449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2